



Summer Olivia Royal

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Education

STANFORD UNIVERSITY, Stanford, CA

- Master of Science: Computer Science, AI concentration | 2026 | GPA: 3.9
- Bachelor of Science: Biomedical Computation | 2025 | GPA: 4.0

Experience

PRODUCT MANAGER: Inspirit AI – Palo Alto, CA (2025-Present)

- Built interactive web applications, Jupyter notebooks, and technical curricula to teach AI concepts
- Lead technical product operations, directly managing 200+ employees, running performance evaluations, allocating work, and advising teams on research direction, technical roadblocks, and client escalations.
- Built python scripts to automate high-volume manual workflows including project matching, creation and archival of project Slack channels, and enrollment of thousands of students across course platforms.
- Served as an on-call software engineer for urgent software bug fixes.

SOFTWARE ENGINEER INTERN: Melio – Mountain View, CA (2024)

- Developed an optimization algorithm for PCR primer selection on a rapid neonatal-sepsis diagnostic platform, maximizing inter-pathogen melt-curve separation while minimizing intra-pathogen variation.
- Built a DNA sequence-alignment and consensus-sequencing pipeline across the platform's pathogen panel to identify phylogenetically similar organisms at elevated risk of diagnostic misclassification.

SOFTWARE ENGINEER INTERN: Immergo Labs – Mountain View, CA (2024)

- Developed immersive virtual environments in Unity for a VR telehealth platform that allows physical therapists to meet with and treat patients remotely.

MEDICAL TECHNOLOGY LEFTEROFF INTERN: Fogarty Innovation– Mountain View, CA (2024)

- Consultations on the biodesign process, regulatory strategy, patents, and medtech commercialization.

SOFTWARE ENGINEER & DATA SCIENCE INTERN: Skywalk – Palo Alto, CA (2023)

- Developed signal-quality algorithms for wrist-worn optical EMG sensors; identified failure modes in an RMS successive-difference detector and designed a variance-based method normalized to optical-signal/power ratio to reduce false positives on non-stationary recordings.
- Designed and executed a 100-participant study to generate ML training data, including a custom Microsoft HoloLens/Unity application for standardized data collection.
- Built python preprocessing pipelines for model-training datasets, automating corrupted-file detection, labeling, and cross-participant standardization.

SOFTWARE ENGINEER INTERN: Argo AI – Palo Alto, CA (2022)

- Built a python tool that extracts diagnostic signals from autonomous vehicle logs and renders them as time-series visualizations with cross-vehicle and cross-signal comparison, time-window filtering, and threshold-based queries, replacing a previously manual workflow.
- Broadened diagnostic tooling access company-wide by removing the prerequisite of python scripting expertise, enabling non-engineering employees to independently analyze and compare log data across multiple vehicles and trips.
- Developed a C++ front-end that unified the safety and diagnostics team's fragmented internal tooling into a single entry point: users pose a diagnostic question in plain language, and the program routes to the appropriate underlying tool, prompts for any missing parameters, and executes it against their data, thus making tools previously known only within the team usable across the organization

Skills

Languages: python, C++, C, Swift, and Java

ML: machine learning, deep learning, LLMs, signal processing, computer vision, data analysis

Frameworks/Platforms: VS Code, Cursor, Unity, Apple HealthKit, Google Colab

Domains: clinical NLP, medical imaging, biosignals, digital health, computational biology

Research

BOUSSARD LAB – Palo Alto, CA (2024 - 2025)

- Developed an end-to-end machine learning pipeline to identify chemotherapy-induced neurotoxicity in a cohort of 27,950 cancer patients, using large language models to extract symptom information from EHR notes and label 100,000+ clinician notes at scale.
- Trained a model to identify, prior to treatment, which patients are at elevated risk of experiencing neurotoxicity so that physicians and patients can make more informed treatment decisions.
- To improve clinical interpretability, I made model cards documenting performance, fairness, and limitations.

DR. JUAN GALEAZZI – Oxford, UK (2023)

While studying abroad at Oxford University, I completed a research tutorial, under the mentorship of Dr. Juan Galeazzi, in computational neuroscience. Topics included brain organization, computational brain modeling, computer vision, memory, reinforcement learning, and large scale neural recordings.

WERNIG LAB – Palo Alto, CA (2021 - 2023)

Studied the reprogramming potential of microglia and the functional integration of induced neurons into neural systems, supporting research into therapeutic approaches for neurodegenerative disease, disease mechanisms of Alzheimer's Disease, and therapeutic methods for neurological patients.

DAVIS & MERCIER LABS – Honolulu, HI (2014 - 2016)

Studied newly discovered fractones (specialized structures in the stem cell niche that bind to neuronal growth factors that control the proliferation and differentiation of stem cells). Our research showed that fractones have been found in the β -amyloid plaques in the brains of Alzheimer's Disease patients.

Side Projects

MIND-CONTROLLED LIGHTBULB

Through Stanford's Brain Computer Interface Club, I created a lightbulb that can be controlled by an individual's thoughts while the individual is wearing an EEG cap; I was responsible for writing the data collection software and signal processing software for this project.

CLIMATE MINDS PROJECT

Developed an app that helps users explore how their values and personal interests are being affected by climate change. I was responsible for training a machine learning model that can continuously process new news articles, filter the factual information, and pull out the cause-and-effect relationships. The long-term goal is for this ML model to be implemented in social media platforms to mark false information.

HIPTRACKS APP

HipTracks is an iOS application leveraging Apple Watch biometric data and artificial intelligence to monitor post-operative hip and knee replacement patients, by using a convolutional neural network classifier to analyze wound photographs and predict the likelihood of surgical site infections. I led front-end development using Swift and integrated backend systems with CardinalKit and Apple HealthKit frameworks.

MEMORY-AS-ACTION FOR RETRIEVAL-AUGMENTED MEDICAL REASONING

- Built a five-stage pipeline (memory bank construction, retrieval, expert annotation, SFT warm-start, GRPO) distilling a 32B "teacher" model into a 7B model that learns to answer USMLE (medical exam) questions
- Ran ablation with self-consistency voting, cloze scoring, and DAgger distillation

BIOMARKER PREDICTION FROM WHOLE-SLIDE HISTOPATHOLOGY IMAGES

- Investigated whether attention-based multiple-instance learning can predict ER, PR, and HER2 biomarker status from H&E-stained whole-slide images as a low-cost alternative to cancer treatment eligibility assays
- Benchmarked three patch encoders (ResNet-50, UNI, CONCH) against three aggregation strategies including a proposed Tumor-Aware CLAM with residual gating, and used patient-level 5-fold cross-validation with bootstrap confidence intervals to establish that dataset size is the binding constraint.

Memory Distillation for Small Scientific Reasoning Models

- Trained small language models to learn when to distrust external memory using supervised fine-tuning and curriculum learning on teacher-generated distillations over a retrieval corpus. Best strategy improved ProtocolQA by 31 points and LitQA2 by 5 points over the 7B zero-shot baseline on LAB-Bench.
- Demonstrated an adversarial limit: targeted surface-form attacks defeat both similarity-based and semantic write-gate defenses by exploiting the same signal retrieval depends on.

REINFORCEMENT LEARNING AGENTS FOR EXPLODING KITTENS

- Built a full-fidelity game simulation (~100,000 hashed states) and three competing agents: MLE with value iteration, Q-Learning with temporal-difference updates, and a Bayesian agent using Dirichlet-Beta inference over deck composition
- Quantified how the game's ~9% per-turn draw risk caps the advantage any policy can extract
- Deployed an interactive website where users play against all three agents

DITCH DAT! HEADLICE PRODUCT

- Created a novel, eco-friendly, affordable, patent-pending pediculicide
- Product won 1st place in Hawaii State Science Fair
- Received federal government funding to prototype further

FOSTER TOWER TREE APP

- Created software app during the COVID pandemic for a residential condominium complex to assist neighbors/residents with daily needs, assistance with shopping/trips to doctors/help with laundry, etc.

Memberships

CO-FOUNDER & FINANCIAL OFFICER

Stanford SupplyHer, a club that fundraises for victims of domestic abuse and other under-resourced women.

GENERAL MEMBER

Tree Hacks, Stanford Mathematical Organization, Women in CS, Stanford Brain-Computer Interfaces, CS for Social Good, Scientists Speak Up, Synapse Brain Injury Support Group, Girl Scouts of America, and Stanford Dance Marathon

Hospital Work

SHADOWING

- Shadowed at Stanford Hospital in the following departments: neurosurgery, neurology, pediatric cardiology, general surgery, and the emergency department
- Shadowed at El Camino Hospital in the following departments: general surgery, gynecology, and radiology

CERTIFICATIONS

John A. Burns School of Medicine, University of Hawaii (2018) Medical Diagnosis & Treatment certification

Honors & Awards

- 1st Place Winner in health{acks} hackathon hosted by Stanford University in the Aging & Longevity Category (2023)
- 1st Place Project Winner for Stanford's Graduate Level Decision Making Under Uncertainty Award (2025)
- Tau Beta Pi member: ranked in the top 10% of GPAs for engineering undergraduates and fulfilled associated community service requirements (2021-2025)
- National Qualifier for Girls Go CyberStart (2020)
- National winner of US Army eCybermission Stem-in-Action Grant (2018)
- 3rd Place International Diamond Challenge Entrepreneur Competition (2020)
- 2nd Place National Genius Olympiad (2020)